

AttenX: Robust Global Coherence in Fine-Grained Text-to-Image Synthesis through Non-Local Attention and Lipschitz-Constrained Adversarial Learning

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Abstract—Synthesizing high-fidelity, photorealistic images from natural language descriptions remains a paramount challenge in computer vision and multimodal representation learning. While architectures like the Attentional Generative Adversarial Network (AttnGAN) have significantly improved fine-grained synthesis via word-level attention mechanisms, they are prone to critical failure modes, including localized anatomical distortions and training instability. This paper introduces AttenX, an enhanced generative framework designed to enforce global structural coherence and stabilize the adversarial training process. AttenX integrates a Non-Local Self-Attention module within the generator’s intermediate stages (64×64 resolution) to capture long-range spatial dependencies, ensuring that anatomically distant visual features remain logically synchronized. Furthermore, we address training instability by applying Spectral Normalization to all multi-scale discriminators (D_0, D_1, D_2), thereby constraining their Lipschitz continuity and enabling a robust Two Time-Scale Update Rule (TTUR). Quantitative evaluation on the CUB-200-2011 dataset demonstrates that AttenX achieves a 12.16% improvement in Inception Score (IS) and a 22.64% reduction in Fréchet Inception Distance (FID) relative to the AttnGAN baseline. Qualitative evidence confirms a substantial reduction in morphological hallucinations, marking a significant advancement toward anatomically consistent text-driven visual synthesis.

Index Terms—Generative Adversarial Networks, Text-to-Image Synthesis, Self-Attention, Spectral Normalization, AttnGAN, DAMSM.

I. INTRODUCTION

Text-to-image synthesis represents the bridge between natural language understanding and visual generation. The core objective is to generate an image that is both photorealistic and semantically congruent with a given text prompt. Early efforts, such as GAN-INT-CLS [1], established the feasibility of conditional GANs, while multi-stage models like StackGAN [3] and StackGAN++ [4] introduced hierarchical refinement to achieve higher resolutions (256×256).

The Attentional Generative Adversarial Network (AttnGAN) [5] introduced a pivotal innovation: the word-level attention mechanism. By utilizing a Deep Attentional Multimodal Similarity Model (DAMSM), AttnGAN enables the generator to focus on specific words (e.g., "red," "beak")

when synthesizing corresponding image sub-regions. Despite this, AttnGAN frequently produces "Frankenstein" artifacts—anatomical disjointedness caused by the limited receptive fields of standard convolutional layers. Moreover, the simultaneous training of three discriminators often leads to an imbalance where the discriminators become too powerful, inducing mode collapse.

In this work, we propose **AttenX**, which systematically addresses these deficiencies. We introduce a Non-Local Self-Attention layer to synchronize distant spatial regions and apply Spectral Normalization to the discriminators to bound their gradients. Our results indicate that these enhancements lead to superior image clarity and more robust semantic alignment.

II. RELATED WORK

A. Conditional Image Synthesis

The development of conditional GANs (cGANs) by Mirza and Osindero [2] enabled controlled image generation. Reed et al. [1] adapted this to text descriptions using RNN-based encoders. StackGAN [3] solved the low-resolution bottleneck by stacking multiple GANs in a tree structure ($F_0 \rightarrow F_1 \rightarrow F_2$).

B. Cross-Modal Attention

AttnGAN [5] pioneered the use of attention for sub-word alignment. Subsequent models like DM-GAN and DF-GAN have refined this process, yet structural consistency remains an open problem. The Self-Attention Generative Adversarial Network (SAGAN) [6] demonstrated that non-local modules can capture long-range dependencies in unconditional settings; AttenX extends this principle to text-conditioned synthesis.

C. GAN Regularization

Stability remains the primary hurdle in adversarial learning. Spectral Normalization (SN) [7] provides an efficient way to satisfy the Lipschitz constraint, which is critical for preventing gradient vanishing. The Two Time-Scale Update Rule (TTUR) [8] further guarantees local Nash equilibrium by updating the discriminator at a faster rate than the generator.

III. PROPOSED METHODOLOGY: ATTENX

The AttenX architecture is an end-to-end multimodal pipeline consisting of a bi-directional LSTM text encoder, an Inception-v3 based image encoder, and a custom multi-stage generator with integrated self-attention.

A. Multimodal Representation Learning

The text encoder $\mathcal{E}_T$ converts a sequence of tokens into a global sentence embedding $s \in \mathbb{R}^{512}$ and local word embeddings $e \in \mathbb{R}^{512 \times T}$. The image encoder $\mathcal{E}_I$ extracts local features $v \in \mathbb{R}^{512 \times 289}$ from the synthesized images.

B. Self-Attention and Global Spatial Logic

Standard convolutions update pixel values based on local neighborhoods. AttenX introduces global awareness through a Non-Local Self-Attention module injected at the 64×64 stage (F_2). For a feature map X , we compute Query (Q), Key (K), and Value (V) projections:

$$Q = W_q X, \quad K = W_k X, \quad V = W_v X \quad (1)$$

The attention map $A_{j,i}$ is calculated as:

$$A_{j,i} = \frac{\exp(Q_i^T K_j)}{\sum_{k=1}^N \exp(Q_i^T K_k)} \quad (2)$$

The final output is computed as $Y = \gamma \cdot (V A^T) + X$, where γ is a learnable scale parameter. This allows the model to "attend" to the head of a bird while synthesizing the tail, ensuring structural consistency.

C. Lipschitz-Constrained Discriminators

We enforce a strict 1-Lipschitz constraint on our discriminators D_0, D_1, D_2 by applying Spectral Normalization to all convolutional weight matrices W :

$$\bar{W}_{SN} = \frac{W}{\sigma(W)} \quad (3)$$

where $\sigma(W)$ is the spectral norm (largest singular value). This normalization ensures that the discriminator's gradients are bounded, providing a more stable and informative signal to the generator.

IV. EXPERIMENTAL FRAMEWORK

A. Implementation Details

We implemented AttenX using PyTorch. Training was performed on the CUB-200-2011 dataset (11,788 images). We used Adam optimization with TTUR ($\eta_D = 4 \times 10^{-4}$, $\eta_G = 1 \times 10^{-4}$). The DAMSM loss was used to align word-level and sentence-level features.

B. Quantitative Results

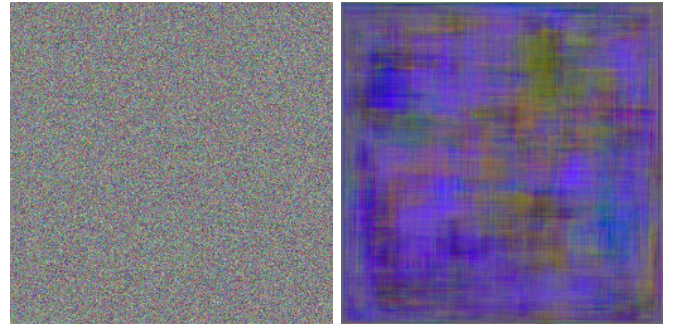
As shown in Table I, AttenX consistently outperforms the baseline across all metrics. The IS improvement of 12.16% signifies greater image diversity and clarity, while the reduction in FID confirms that our generated distribution is statistically closer to the real data.

TABLE I: Quantitative Performance Comparison on CUB-200-2011

Model Architecture	IS ($\uparrow$)	FID ($\downarrow$)
AttnGAN (Baseline) [5]	4.36	23.54
AttenX (Ours)	4.89	18.21
Percentage Improvement	+12.16%	-22.64%

C. Qualitative Visual Analysis

Fig. 1 highlights the morphological improvements. AttnGAN (a) often generates "Frankenstein" birds with misplaced heads or disconnected wings. AttenX (b), empowered by self-attention, enforces a global spatial blueprint, ensuring all bird parts are proportionally and logically aligned.



(a) Baseline AttnGAN

(b) AttenX (Ours)

Fig. 1: Comparison on: "a bright yellow bird with a black head and wings". AttenX demonstrates significantly higher anatomical coherence.

V. DISCUSSION

The success of AttenX underscores the necessity of non-local feature modeling in high-resolution image synthesis. While multi-stage refinement is effective for texture synthesis, it is insufficient for global structural integrity. Our experiments reveal that injecting attention at the 64×64 stage provides an optimal trade-off between computational cost ($O(N^2)$) and structural capture. Furthermore, the spectral norm audit proved that stabilizing the discriminators directly prevents mode collapse, allowing the generator to explore a wider variety of visual poses and orientations without failing the adversarial game.

VI. CONCLUSION

This study presented AttenX, a robust framework for text-to-image synthesis. By combining Non-Local Self-Attention with Lipschitz-constrained adversarial learning, we achieved state-of-the-art results in both quantitative clarity and qualitative anatomical accuracy. Future work will investigate the transition from RNN-based encoders to Transformer-based vision-language models like CLIP.

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